Amir Besher

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Education

Bachelor's Degree in Computer Science **08/2023 - 05/2027**
American University of Ras Al Khaimah (AURAK) Ras Al Khaimah, UAE
CGPA: 3.8 **GRE General Test (10/2026):** Verbal 159 (79th pct.), Quantitative 163 (60th pct.), Analytical Writing 3.5

Modules: Artificial Intelligence Data Science Statistics Linear Algebra Discrete Mathematics Parallel Programming Design & Analysis of Algorithms OOP

Research Experience

Research Assistant **08/2026 - Present**
Advanced Technology and Artificial Intelligence Center Ras Al Khaimah, UAE
Supervisor: Prof. Khouloud Salameh (Associate Professor of Computer Science)

Checkpoint Early-Warning Prediction of Student Performance **09/2026 - Present**

- Designed and built a system that estimates each student's risk of failing or withdrawing, final letter grade and term GPA at weeks 0, 4, 8, 12 and 16 of a term, using only information available at each checkpoint (final exams excluded; coursework scores counted only once released).
- Compared explicit baselines, interpretable models (elastic net, proportional-odds and L2 logistic regression) and LightGBM under term-forward splits, selecting a model only if it showed skill on a separate calibration term.
- Calibrated probabilities (Platt, isotonic and temperature scaling) and added conformalized prediction intervals, advisor-capacity risk tiers and a subgroup fairness audit (recall, calibration gap, ABROCA).
- Evaluated on three public datasets (OULAD, KU Leuven ALPACAS, SED). On OULAD, PR-AUC rose from 0.58 at week 0 to 0.77 at week 16; documented interval under-coverage and calibration loss when the test term differed from the training terms.
- Built a local FastAPI prediction service and an accessible advisor dashboard, and contributed the model comparison and study review tables to the team's systematic literature review of student performance prediction.

Automatic Similarity Weighting for Dental X-ray SVG Annotation **08/2026 - 09/2026 (paused)**

- Studied SSG (Salameh et al., *Multimedia Systems*, 2023), an unsupervised framework that annotates panoramic dental X-ray images in SVG form by combining shape, area, position and colour similarity with user-set weights.
- Reviewed literature on selecting these weights automatically (grid and random search, genetic algorithms, particle swarm and Bayesian optimization), with annotation F1 as the objective, to replace manual weighting. The project resumes in Spring 2027.

Industry Experience

Mobile Planning Intern **05/2026 - 07/2026**
e& (Etisalat) Abu Dhabi, UAE

- Studied 5G network planning principles, focusing on coverage prediction, capacity planning and performance optimization.
- Evaluated antenna deployment strategies and radio propagation factors affecting network coverage and quality of service.
- Produced technical reports and presentations communicating engineering analyses and project findings to supervisors.

Network & Security Technologies Intern **05/2025 - 07/2025**
Alpha Data Abu Dhabi, UAE

- Monitored enterprise infrastructure using OpManager, Log360, Microsoft Defender, and LogRhythm SIEM to support network performance and security operations.
- Analyzed security alerts, system logs and incident data to identify potential threats and support cybersecurity investigations.
- Collaborated with security and network engineers to investigate operational issues, monitor infrastructure health and respond to security events.

Selected Projects | Full Portfolio: amirbeshir.me

AURAK Research Grants Dashboard (University Software Competition) **09/2026 - Present**

Project: Building a dashboard that finds research grants, fellowships and scholarships for AURAK students and faculty and matches each opportunity to faculty whose research fits its scope.

- Integrated the Grants.gov and EU Funding & Tenders APIs and UAE funder scrapers in a FastAPI, SQLite and Next.js stack.
- Matched opportunities to 96 faculty using sentence-transformer embeddings, FAISS and OpenAlex publication records.

- Each match shows its evidence and is reported as a match score, not a probability, with no forced match when nobody fits.

Student At-Risk Recovery Planner using A* Search **03/2026 - 05/2026**

Project: Developed an AI-powered decision support system that generates personalized academic recovery plans using the A* search algorithm with custom cost functions to identify optimal pathways toward graduation.

- Implemented the A* algorithm from scratch, designing domain-specific heuristic functions and action costs to efficiently explore recovery strategies.
- Built an interactive Python dashboard for scenario analysis to evaluate recovery plans under different academic constraints.
- Compared A* search against GBFS and UCS algorithms to evaluate solution quality and search efficiency.

Parallel Bank Transactions and Fraud Detection Simulation **03/2026 - 05/2026**

Project: Developed a high-performance transaction processing simulator to evaluate serial, multithreaded, and multiprocessing architectures while detecting suspicious transaction patterns.

- Implemented serial, thread and process-based transaction processing systems using Python's threading and multiprocessing libraries.
- Benchmarked execution time, speedup, and scalability across multiple processing configurations.
- Designed fraud detection rules to identify suspicious transaction patterns during concurrent execution.
- Visualized performance metrics and analyzed synchronization challenges including race conditions and resource contention.

Network Analysis of Human Apoptosis Protein Interactions **04/2025 - 05/2025**

Project: Applied graph theory and network analysis techniques to investigate the structural properties of human apoptosis protein-protein interaction networks using computational methods. **Code: Protein Interactions**

- Constructed and analyzed biological interaction networks using Python and NetworkX.
- Identified influential proteins through graph-theoretic analysis and visualized interaction patterns via network visualizations.
- Interpreted characteristics of biological networks to demonstrate the application of computational methods in bioinformatics.

Leadership and Outreach

Student Government Association (SGA) Treasurer **08/2026 - Present**
American University of Ras Al Khaimah Ras Al Khaimah, UAE

- Managed SGA finances, including budgeting, fund allocation and account reconciliation, ensuring transparency.
- Maintained financial records, expenditure reports and audit documentation for student organizations and committees.
- Served on the Executive Board, presenting financial updates and resource plans to the Board and university administration.

Key Skills

Programming:	• Python	• C++	• Java	• SQL	• LaTeX
ML & Data Libraries:	• scikit-learn	• LightGBM	• pandas	• NumPy	• NetworkX
Tools & Technologies:	• VS Code	• Git	• MySQL	• FastAPI	• NI Multisim
Languages:	• English: Fluent	• Arabic: Native			

Computer Science Skills:

• Machine Learning	• Model Evaluation	• Optimization	• Algorithm Analysis
• Parallel Computing	• Data Structures	• Performance Benchmarking	• Operating Systems
• Cybersecurity	• Network Security	• Digital Logic	• Research

Soft Skills:

• Time Management	• Communication	• Adaptability	• Problem-solving	• Leadership
• Attention to Detail	• Critical Thinking	• Work Ethic	• Independence	• Creativity

Honors, Awards, and Recognition

- President's List (GPA 4.0/4.0) awarded for achieving a perfect academic record in Spring 2026.
- Dean's List (Fall 2023, Fall 2024, Spring 2025, Fall 2025) for outstanding academic performance.
- Merit-based scholarships awarded by AURAK: 60% (2026), 50% tuition waiver (2025) and 40% (2023, 2024).
- Certificates of Completion: Advanced Algorithms and Complexity, UC San Diego via Coursera (2026), AI Technology and Applications, Huawei ICT Academy (2026), Operating Systems, BITS Pilani via Coursera (2026), and HCIA-Security V4.0, Huawei ICT Academy (2025).